



Can you find an equation giving the level of caffeine in the blood after an amount of coffee. There should be some demographical data to enter. Also I want some indication of if that level is high. Especially what level interrupts sleep

You can approximate blood caffeine over time after coffee with a one-compartment first-order elimination model, then plug in demographic modifiers (age, pregnancy, smoking, OCP use, etc.) via the elimination half-life and volume of distribution. ^[1]

Below is a clinically usable framework rather than a perfect PK model, plus a rule-of-thumb threshold for when sleep is likely to be disturbed.

Basic pharmacokinetic model

Assume instant absorption (or a short absorption phase) and first-order elimination. ^[1]

For a single dose of caffeine D (mg), volume of distribution V_d (L), and elimination half-life $t_{1/2}$ (h), plasma concentration over time is:

$$C(t) = \frac{D}{V_d} \cdot e^{-kt}, \quad k = \frac{\ln(2)}{t_{1/2}}$$

This is a standard one-compartment IV bolus model; for oral caffeine this is a reasonable approximation because absorption is rapid (peak plasma at 30–75 min). ^{[2] [1]}

For multiple cups, sum across all doses:

$$C(t) = \sum_{i=1}^n \frac{D_i}{V_d} \cdot e^{-k(t-t_i)} \quad \text{for } t \geq t_i$$

where t_i is the time of cup i .

You can implement this easily in R for arbitrary intake patterns.

Demographic inputs to the equation

You can let the user enter:

- Weight (kg).
- Age group (e.g., adult <65 vs ≥65).
- Sex and oral contraceptive use.
- Pregnancy status.
- Smoking status.
- Severe liver disease (yes/no).
- Usual caffeine use (to flag tolerance, not PK).

These map to PK parameters, mainly $t_{1/2}$ and V_d .

1. Volume of distribution

Total body water is a decent proxy. Caffeine V_d is typically about 0.6 L/kg. ^[1]

So:

$$V_d \approx 0.6 \times \text{weight (kg)} \quad [\text{L}]$$

Example: 70 kg $\rightarrow V_d \approx 42$ L $\rightarrow$ a 100 mg bolus gives peak $C(0) \approx 100/42 \approx 2.4$ mg/L.

2. Half-life adjustments

Baseline adult non-smoker, not pregnant, no OCP, no cirrhosis:

- $t_{1/2} \approx 4\text{--}5$ h is typical. ^[1]

Modifiers (directionally, approximate factors; values vary between studies):

- Cigarette smoking: shorter half-life, e.g. 3–4 h ($\approx 0.7\text{--}0.8\times$). ^[1]
- Oral contraceptives: half-life can roughly double to around 8–10 h. ^[1]
- Late pregnancy: half-life can increase up to $\approx 10\text{--}18$ h. ^[3] ^[1]
- Severe liver disease: markedly prolonged; often >10 h. ^[1]
- Older age: some reports of modest prolongation (e.g. +20–30%), but data are heterogeneous. ^[1]

You can encode this as:

$$t_{1/2} = t_{1/2,\text{base}} \times f_{\text{smoke}} \times f_{\text{OCP}} \times f_{\text{preg}} \times f_{\text{liver}} \times f_{\text{age}}$$

with, for example:

- $t_{1/2,\text{base}} = 4.5$ h
- $f_{\text{smoke}} = 0.7$ if smoker, else 1.0
- $f_{\text{OCP}} = 2.0$ if on OCP, else 1.0

- $f_{\text{preg}} = 2.0$ if 3rd trimester, else 1.0
- $f_{\text{liver}} = 2.0$ if significant hepatic impairment, else 1.0
- $f_{\text{age}} = 1.3$ if ≥ 65 y, else 1.0

The exact factors can be tuned if you want to tie them tightly to specific studies, but this gives a usable clinical calculator.

3. Example equation with demographics

Putting it together for one cup of coffee with dose D at time t_0 :

$$C(t) = \frac{D}{0.6 \times \text{weight}} \cdot \exp \left(- \frac{\ln 2}{4.5 \cdot f_{\text{smoke}} \cdot f_{\text{OCP}} \cdot f_{\text{preg}} \cdot f_{\text{liver}} \cdot f_{\text{age}}} (t - t_0) \right)$$

For multiple cups, sum across doses as above.

Dose per “cup” can be user-defined; brewed coffee typically ranges ~70–150 mg per 150–240 mL depending on preparation.^{[4] [2]}

When is the level “high”?

There is no universally agreed plasma threshold for “high,” but some anchors:

- “Normal” morning coffee drinker with 1–3 cups will often sit in the low single-digit mg/L range at peak (roughly 2–10 mg/L depending on dose and timing).^[1]
- Toxicity is generally reported at much higher levels (e.g. tens of mg/L and above in overdose contexts).^[1]

For everyday guidance, you might define:

- Low: <2 mg/L (likely minimal effect in a tolerant adult).
- Moderate: 2–10 mg/L (typical of habitual use, clearly psychoactive).
- High: >10 mg/L (unusual outside heavy use or slow metabolism; more side effects likely).

These cut-points are pragmatically chosen for a calculator rather than evidence-based thresholds, but sit within usual PK ranges.^[1]

Levels and sleep disruption

Experimental data on timing

A double-blind crossover study (400 mg caffeine, ~moderate-high dose) gave caffeine at 0, 3, or 6 h before bedtime and found significant sleep disturbance in all three conditions.^{[5] [6]}

- Total sleep time was reduced, sleep efficiency decreased, and sleep onset latency increased even when caffeine was given 6 h pre-bedtime.^{[6] [5]}

- This supports the common recommendation to avoid substantial caffeine at least 6 h before planned sleep. ^{[7] [5]}

Linking to plasma levels

With a 4.5 h half-life, after 6 h:

$$C(6) \approx C_{\text{peak}} \cdot 2^{-6/4.5} \approx C_{\text{peak}} \cdot 0.39$$

So about 40% of peak concentrations remain 6 h after intake in an average metabolizer. ^[1]

Using the example above (70 kg, 150 mg dose):

- $C_{\text{peak}} \approx 150/42 \approx 3.6$ mg/L
- At 6 h: $C(6) \approx 1.4$ mg/L

Yet 400 mg taken 6 h before bed still significantly disrupts sleep, so sleep is clearly affected at concentrations in the low mg/L range in controlled conditions. ^{[5] [6] [2]}

Given this, a pragmatic “sleep-risk” categorization for a calculator:

- Probably minimal impact for most: <1 mg/L at bedtime.
- Possible impact, especially in sensitive, anxious, or insomnia-prone individuals: 1–2 mg/L at bedtime.
- Likely impact for many: >2 mg/L at bedtime.

These thresholds are not formally validated but are consistent with the PK of typical doses and the 6-h rule from experimental data. ^{[6] [2] [5]}

Individual susceptibility

A classic small study comparing people with and without “coffee-attributed insomnia” found that those who reported insomnia had longer plasma half-life (≈ 7.4 vs 4.2 h) and lower clearance. ^[8]

- They had higher midnight plasma caffeine after afternoon coffee, and reported insomnia more often. ^[8]
- This supports using a longer half-life factor in people who report marked caffeine sensitivity.

How you might implement this in R

Given your background, I’d suggest:

1. Ask user for:
 - Body weight, sex, age group, pregnancy, smoking, OCP, liver disease.
 - Bedtime.
 - Times and doses of all caffeine intakes (coffee, tea, soda, pills).

2. Derive V_d and $t_{1/2}$ as above.

3. Create a function:

```
caffeine_conc <- function(t, dose_times, doses, weight, t_half) {  
  Vd <- 0.6 * weight  
  k <- log(2) / t_half  
  conc <- sapply(t, function(tt) {  
    sum(doses * exp(-k * pmax(0, tt - dose_times))) / Vd  
  })  
  conc  
}
```

4. Evaluate at bedtime (and across the day for a plot) to give:

- Caffeine level at bedtime (mg/L).
- A qualitative label: “low”, “moderate”, “high”, plus “likely to disturb sleep: yes/no/uncertain” based on the thresholds above.

Clinical interpretation summary

- Use a one-compartment, first-order elimination model with demographic modifiers on $t_{1/2}$ and weight-scaled V_d .^[1]
- Most adults with normal metabolism will have ~50% of peak caffeine remaining after one half-life, ~25% after two, etc., so late-afternoon coffee can leave meaningful levels at bedtime.^{[2] [1]}
- Experimental data show that 400 mg caffeine can significantly disrupt sleep even when taken 6 h before bedtime, implying that relatively low mg/L concentrations at bedtime can be problematic.^{[5] [6]}

Would you like me to help you turn this into a small R function or Shiny app where you can enter dose times, demographic variables, and get both a time-curve and a “sleep risk” flag?

✱

1. <https://www.ncbi.nlm.nih.gov/books/NBK223808/>
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